



Home health hub block diagram

medical end-devices connect wirelessly via the personal area network to an application hosting device, such as mobile phone, PDA, PC, tablet PC or set-top box. This application hosting device then uses a wired (Ethernet) or wireless (cellular) signal to transmit the data to a storage location such as an electronic medical record or personal medical record.

Over time, other more unique form factors that combine multiple consumer medical devices with an application hosting device, like a medical kiosk that is used within hospitals, doctor's offices and pharmacies, will become prevalent.

Safety. Safety can be enhanced through use of the latest MCUs and sensors. A good example of how technology has improved user safety is in the field of blood-glucose testing for diabetes management.

Years ago, diabetic patients had their blood tested at a doctor's office every week or so. This provided an incomplete picture of the patient's glucose level, allowing only coarse control. The advent of portable self-testing glucose meters allows patients to test their blood sugar several times a day, allowing much finer control of glucose levels. At the same time, the transition to electrochemical sensors interfaced to precision analogue components has led to much more accurate readings.

In many ways, miniaturisation actually eases the problem of safety. For example, in many applications, there is a requirement to maintain 5kV isolation

between the patient and the main power outlet. As a portable device is often battery-operated, it provides inherent isolation. However, care must still be taken in the system design—for example, when wired communication such as USB is used. Thankfully, help is at hand—semiconductor companies supply USB isolators that comply with the medical standards.

Data security and privacy. The communication link gives an opportunity for the data to be intercepted by an unauthorised party. If the data being transmitted is sensitive medical information, this is clearly unacceptable.

While it is relatively easy to ensure privacy with a wired USB connection, the situation is less robust with wireless connections. Every day, lots of medical information about patients moves through cyberspace, which makes it



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vulnerable to corruption or interception.

These issues have been comprehensively addressed in standard wireless communication protocols such as Bluetooth. Standards groups such as Continua Alliance have done great work to ensure that data integrity is maintained throughout the complete data chain from portable medical device to hospital and personal health records stored in the cloud.

With increased risk of exposure to threats from cyber-criminals, data encryption is required to establish a secure connection between the provider and the patient.

Data encryption algorithms such as the KEELQ advanced code-hopping technology can protect data during transmission by rendering it unreadable to unauthorised receivers. When secure encryption is included in the programming of MCUs in distributed data systems, every link in the data chain can be secured.

Experts argue that the most secure connection will be based upon dynamic keying. A direct virtual private network connection will be established based upon continuous rapid sampling of the data stream to change the code for both the sender and the receiver.

Reliability. Compliance to quality system standards such as ISO/TS-16949 is important for a semiconductor vendor who supplies products to a medical device manufacturer. Microchip's adherence to the ISO/TS-16949 quality standards aligns well with the requirements of medical device manufacturers for stability and high reliability.

Reliability of the electronics part of portable equipment is generally orders of magnitude higher than other parts of the equipment, such as software algorithms or body contact. Take, for example, fall detectors used for elderly patients. The most common source of a false positive reading is due to algorithmic interpretation rather than raw readings from an accelerometer. Similarly, with ECG measurement systems, an incorrect result is most likely to be caused by poor reliability of the electrode-body contact, rather than